



Water, soil, and nutrient losses related to surface runoff in conventional and conservationist horticultural systems

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Abstract

Purpose This study aimed to quantify the losses of water, soil, and nutrients through surface runoff during natural rainfall events and to evaluate different soil management systems for garlic cultivation.

Methods The experiment was conducted using a clayey Cambisol in 2023 and 2024. The crop sequence included cover crops before garlic during autumn/winter, followed by common beans in summer. The treatments included were as follows: (i) CT: conventional tillage with soil disturbance and no cover crops before garlic; (ii) BO: garlic over black oat residue; (iii) FR: garlic over forage radish; and (iv) WP: garlic over winter pea, with the three cover crops cultivated under a no-tillage (NT) system. The galvanized plates delimited the runoff area (4.25 m²), directing surface flow to the collectors. After each rainfall event, the losses of water, soil, total nitrogen (N_T), total organic carbon (TOC), and soluble and total phosphorus (P_{SOL} and P_T, respectively) were determined.

Results Water loss was 127% greater in 2023, with losses of N_T, TOC, and P_{SOL} being 84%, 21%, and 155% higher, respectively, than in 2024. Compared with the CT system, the NT system reduced soil loss by approximately 46%, regardless of the cover crop used. The greatest water, soil, and nutrient losses occurred in garlic (94%) in 2023 and in beans (74%) in 2024. The P_T concentration in the runoff was 22 and 66 times higher than the limit specified under environmental legislation in 2023 and 2024, respectively.

Conclusion Cover crops effectively reduced soil losses but were less effective at mitigating water and nutrient losses through surface runoff.

Highlights

- Annual climatic conditions are more relevant than crop management in nutrient transport by surface runoff in horticultural systems
- Cover crops reduce soil erosion but do not eliminate nutrient loss
- Soil preparation for garlic cultivation significantly increases soil loss
- Garlic yield is not negatively affected by the adoption of no-tillage farming

Keywords Conventional tillage · Soil erosion · Eutrophication · Cover crops · Soil protection

1 Introduction

The primary cultivation of vegetables covers 59.6 million hectares worldwide (FAOSTAT 2026). Garlic (*Allium sativum* L.) is among the most widely cultivated vegetables worldwide, with production reaching 29.7 million tons in 2024 (FAO and CIRAD 2021). In Brazil, approximately 12.9 thousand hectares are harvested (IBGE 2026), predominantly using conventional tillage (CT), which involves ploughing, harrowing, and rotary hoeing to form beds (Hahn et al. 2024b). This intensive management approach can compromise soil structure, promoting surface runoff and erosion (Carretta et al. 2021), particularly in areas lacking soil cover or conservation practices (Barbosa et al. 2021). Furthermore, it accelerates the loss of nutrients and carbon, particularly in sloping areas exposed to high-intensity rainfall (Bertol et al. 2007; Schick et al. 2014). These losses can lead to the accumulation of chemical elements in surface waters at concentrations that exceed established safety thresholds, resulting in eutrophication (Dodds et al. 1998; Grando et al. 2021).

Horticultural crops such as garlic leave little residue on the soil surface after harvest. In many regions, garlic is grown in succession with common beans or soybeans, which also provide minimal straw cover owing to their low C/N ratio (Hahn et al. 2024b). Researchers have reported that beans produced only 2.4 Mg ha⁻¹ of residue, which is insufficient to maintain adequate soil protection. Combined with high annual precipitation, this scenario intensifies the direct impact of raindrops, causing soil pore clogging and compromising soil physical quality (Pessotto et al. 2021). This reduces infiltration due to the destruction of soil aggregates and reduction in macroporosity, which facilitates water infiltration (Rauber et al. 2026), thereby increasing surface runoff volume and consequently exacerbating erosion (Figueiredo et al. 2025).

Garlic cultivation requires high rates of fertilizer, with applications reaching up to 350 kg ha⁻¹ of N and 570 kg ha⁻¹ of P₂O₅ (CQFS-RS/SC 2016). Some of these nutrients may be lost through surface runoff, particularly N (Bouraima et al. 2016) and P (Grando et al. 2021), which increases the risk of eutrophication of water bodies. Even in conservation systems, higher nutrient concentrations in the surface layers can intensify these losses (Grando et al. 2023). In this context, a no-tillage (NT) vegetation system based on continuous soil cover, crop rotation, and minimal soil disturbance is needed (Hahn et al. 2024b). This conservation management approach has already produced positive results for various crops, including tomato (Samedani and Meighani 2022), chayote, cabbage, zucchini (Mauri et al. 2024), broccoli (Ferreira Junior et al. 2023), and onion (Câmara et al. 2026; Rauber et al. 2026). However, implementing an NT system for garlic remains challenging because of a lack

of specialized machinery for sowing garlic cloves and the limited availability of cover crops that can be grown before garlic is planted without competing with summer crops such as soybeans or beans.

Off-season cover crops are fundamental to this system, as they provide soil protection (Schick et al. 2017) and represent an alternative means of increasing annual biomass inputs (Siqueira et al. 2025). Grasses such as black oat can accumulate large amounts of biomass and have a high C/N ratio, providing more durable soil cover; however, they may temporarily immobilize N (Ferreira et al. 2014; Tassinari et al. 2021). Legumes such as vetch (*Vicia sativa*) and winter pea (*Pisum sativum*) have low C/N ratios, allowing rapid N release via biological fixation (Kehoe et al. 2024). Conversely, crucifers such as forage radish (*Raphanus sativus*) promote nutrient cycling and soil decompaction (Siqueira et al. 2025), although they produce less persistent cover. Therefore, the choice of cover crops must balance climatic conditions with the synchronization of soil protection and nutrient cycling according to the main crop to increase the efficiency of the NT system.

Although NT farming is well established for grain crops, few studies have focused on vegetables such as garlic, which require low levels of organic residue. Garlic is also often cultivated in regions with steep slopes and heavy rainfall. The development of sustainable horticulture is crucial for creating resilient food systems, reducing negative environmental impacts, and addressing the challenges posed by climate change (Hanif 2024). In light of the limitations of CT and the advantages of conservation systems, evaluating the efficiency of autumn/winter cover crops prior to garlic planting is essential for reducing erosion. The innovation of this study lies in the generation of information on the NT system for garlic cultivation, the measurement of losses, and the assessment of the effects of cover crops on erosion control. This approach is in line with the 17 Sustainable Development Goals (SDGs) of the 2030 Agenda (United Nations 2015). We hypothesize that management practices differ in their effectiveness in reducing soil and nutrient losses. Thus, this study aimed to compare and quantify annual water, soil, and nutrient losses through surface runoff across different soil management systems for garlic cultivation.

2 Materials and methods

2.1 Location and characterization of the experimental area

The experiment was conducted between January 2023 and December 2024 in Curitiba, Santa Catarina (27°16'42" S and 50°30'03" W, altitude of 998 m), southern Brazil

(Fig. 1). According to the Köppen climate classification, the region has a Cfb climate, which is humid subtropical with mild summers (Alvares et al. 2013). The average annual maximum temperature is 22.0 °C, the minimum temperature is 12.4 °C, and the average annual precipitation is 1,678 mm (Data 2025; Wrege et al. 2012).

The soil in the experimental area was classified as Cambissolo Háplico (SiBCS 2025), which corresponds to a Cambisol (WRB 2022). Before the experiment was implemented, the 0.00–0.20 m soil layer exhibited the following characteristics: 590 g kg⁻¹ clay (pipette method) and 32.9 g kg⁻¹ organic matter (Walkley-Black); a pH (H₂O) of 5.3

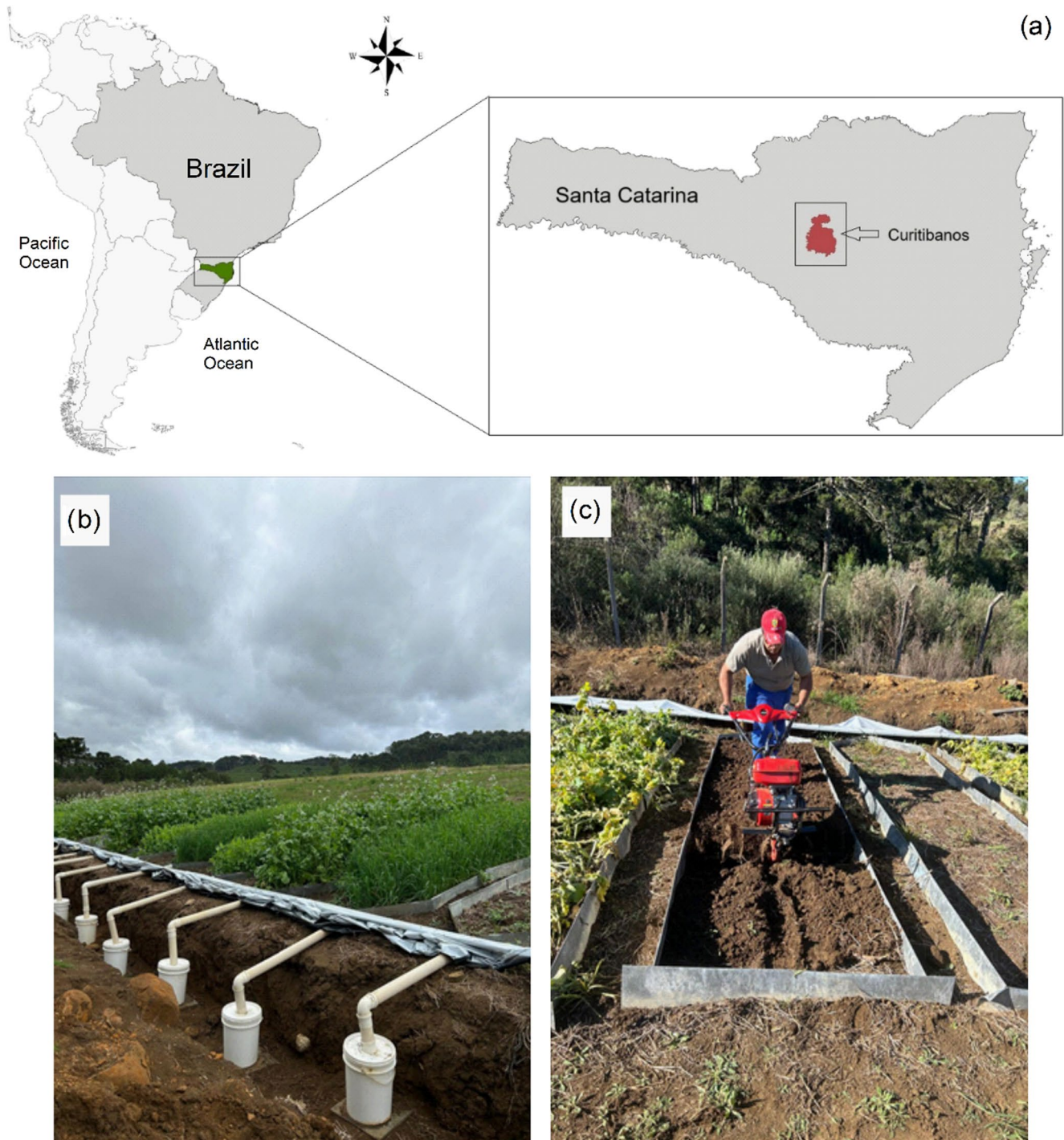


Fig. 1 Experimental location in the municipality of Curitiba, Santa Catarina state, southern Brazil (a); experimental units constructed with galvanized plates to direct runoff water into collection buckets (b); and

soil in the conventional tillage (CT) plot being tilled with a power tiller before garlic planting (c)

(1:1 soil-to-water ratio); Ca and Mg contents of 4.1 and 3.4 $\text{cmol}_c \text{ kg}^{-1}$, respectively (extracted with 1.0 mol L^{-1} KCl); and available P and K concentrations of 7.6 and 56.3 mg kg^{-1} , respectively (extracted with Mehlich-1) (Tedesco et al. 1995). Until 2013, the area was cultivated with pine (*Pinus taeda*). After pine was harvested, soil fertility was restored by applying limestone and chemical fertilizers according to the CQFS-RS/SC (2016) recommendation, and grain cultivation began using NT systems with maize (*Zea mays*) or soybean (*Glycine max*) in the summer and black oat (*Avena strigosa*) in the winter.

2.2 Experimental design

In December 2022, the area was cultivated with common beans (*Phaseolus vulgaris*), with the seeds sown perpendicular to the slope. After bean was harvested in January 2023, experimental plots were established in a randomized complete block design with four replicates. Each plot was defined by the insertion of 4.25-m-long galvanized plates in the direction of the slope, with a width of 1.0 m (Fig. 1), totaling 4.25 m^2 . A funnel was installed at the lowest end of each plot and connected to a 20-L collector to store the surface runoff.

The experiment was conducted on a 13% slope and involved evaluation of soil management and cover systems for garlic (*Allium sativum* L.) cultivation. The treatments

included were as follows: (i) CT: conventional tillage with bed formation and no cover crops prior to garlic cultivation (Fig. 1c); (ii) BO: garlic over black oat (*Avena strigosa*) straw; (iii) FR: garlic over forage radish (*Raphanus sativus* L.) straw; and (iv) WP: garlic over winter pea (*Pisum sativum* L.) straw, with the three cover crops cultivated under an NT system (Fig. 2).

The cover crops were sown in April 2023 and 2024. The sowing density was 70 kg ha^{-1} for black oat, 17 kg ha^{-1} for forage radish, and 64 kg ha^{-1} for winter pea, with a spacing of 0.20 m maintained between rows. The cover crops were fertilized according to soil analysis and regional fertilization recommendations. Only 90 kg ha^{-1} P_2O_5 and 90 kg ha^{-1} K_2O were applied in 2023 (CQFS-RS/SC 2016). The cover crops were not fertilized in 2024 because of high concentrations of nutrients in the soil, in accordance with the technical criteria of CQFS-RS/SC (2016).

In July of both years, the cover crops were cut close to the ground within a 0.5 m^2 area of each experimental plot to estimate biomass production. The crops were then desiccated (a.i. glyphosate, 3 L ha^{-1}). Seven days after desiccation, garlic was planted in July of each year. Under the CT system, the soil was tilled with a power tiller at the time of garlic planting (Fig. 1c). In the cover crop treatments, soil was not disturbed, with cultural residues maintained on the soil surface and planting furrows opened manually for the garlic.

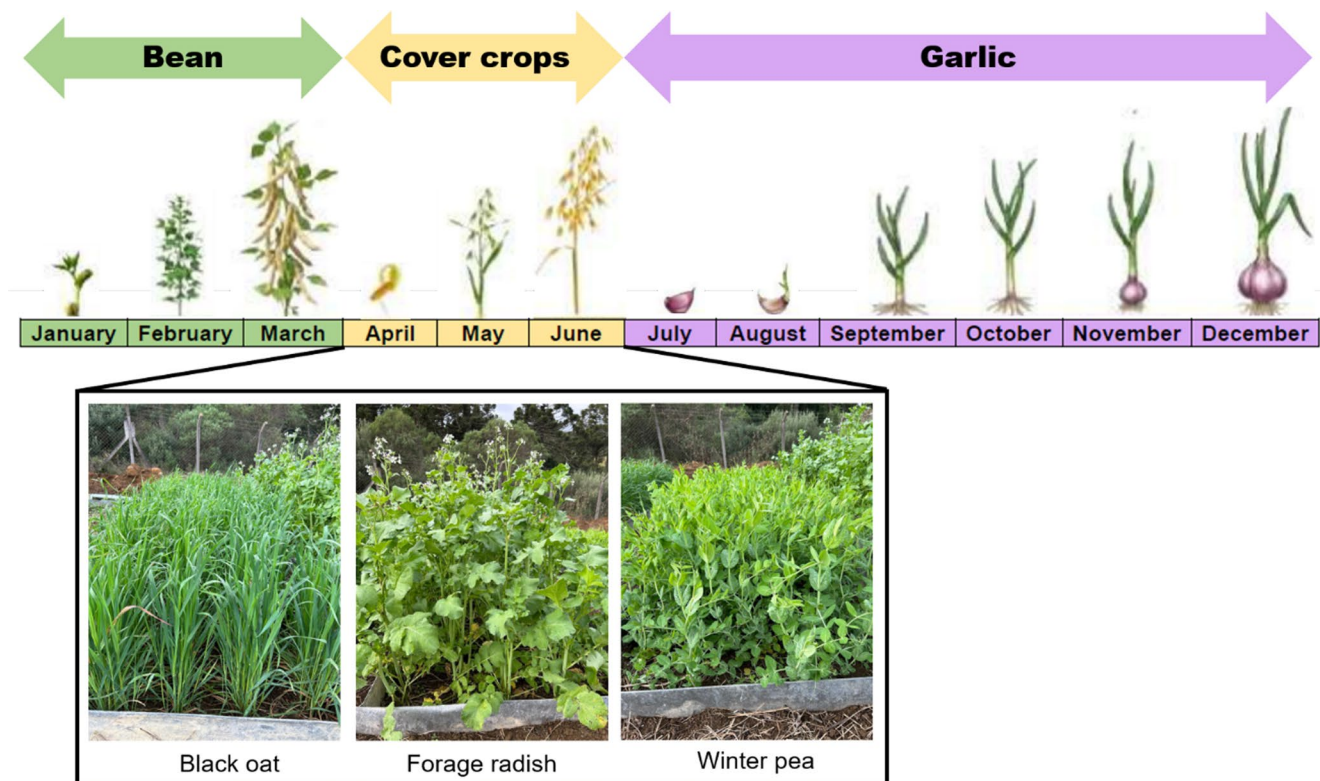


Fig. 2 Schematic representation of the crops grown during the 2023 and 2024 agricultural years

The garlic cultivar used was ‘San Valentim’, which belongs to the ‘noble’ group and produces bulbs classified as types 6 (diameter 47–56 mm) and 7 (diameter > 56 mm) (Resende et al. 2013). The plants were grown in single rows (four rows per bed), with 0.20 m spacing maintained between rows and 0.10 m between plants, resulting in a total population of 400,000 plants ha⁻¹. The cultivation periods throughout each agricultural year are shown in Fig. 2.

Basal fertilizer was incorporated into the entire bed under the CT system and applied to the furrow line under the NT system. In 2023, the basal fertilizer rate applied was 100 kg ha⁻¹ of N, 435 kg ha⁻¹ of P₂O₅, and 440 kg ha⁻¹ of K₂O (P and K correction and maintenance rates). In 2024, the basal fertilizer rate applied was 100 kg ha⁻¹ of N, 200 kg ha⁻¹ of P₂O₅, and 200 kg ha⁻¹ of K₂O (P and K maintenance rates). In both years, topdressing was applied at a rate of 100 kg ha⁻¹ of N as urea, with each application performed at 25 days after planting and 15 days after garlic differentiation. Fungicides and insecticides registered for garlic cultivation were applied for phytosanitary control. ‘Chapecó’ was the common bean cultivar used at a density of 250,000 plants per hectare (0.45 m spacing between rows x 0.09 m between plants). In 2023, 50, 80, and 80 kg ha⁻¹ N, P₂O₅, and K₂O, respectively, were applied. No fertilizer was applied in 2024 because of the high concentrations of nutrients in the soil due to garlic fertilization.

2.3 Collection and analysis of water, soil, and nutrient losses

Water, soil, and nutrient losses were evaluated following rainfall events that caused surface runoff throughout the growing cycles. A total of twenty runoff events occurred in 2023 and nine in 2024. The runoff water was collected in containers, and its total volume was determined using a graduated cylinder. A 250-mL sample was sent to the laboratory for soil quantification and determination of the concentrations of total organic carbon (TOC), total nitrogen (N_T), soluble phosphorus (P_{SOL}), and total phosphorus (P_T).

Soil quantification (g L⁻¹) was performed by drying a 50-mL sample of runoff in an oven at 65 °C. Soil TOC was determined using the oxidation–reduction method (Embrapa 2017). Soluble P (P_{SOL}) in an aqueous solution was determined by adding ammonium molybdate to an acidic medium (APHA 2017). For N_T and P_T analyses, the samples were subjected to sulfuric acid digestion (H₂SO₄+H₂O₂) following the protocol of Tedesco et al. (1995), which is summarized as follows: 5 mL of the sample was added to 1 mL of hydrogen peroxide (H₂O₂), 2 mL of sulfuric acid (H₂SO₄), and 0.7 g of the digestion mixture. After digestion, the samples were diluted to 25 mL. N_T was determined using the Kjeldahl distillation method, whereby 15 mL of sodium hydroxide (NaOH) was added to an aliquot from the

sulfuric acid digestion, and the distillate was collected in a boric acid (H₃BO₃) solution. The distillate was then titrated with H₂SO₄. All extracts (P_{SOL} and P_T) were analyzed using a molecular absorption spectrophotometer at 882 nm, according to the method of Murphy and Riley (1962).

2.4 Soil collection

Soil samples were collected after the garlic harvest each year to evaluate the chemical composition of the 0.00–0.10-m and 0.10–0.20-m soil layers. The samples were dried in a forced-air circulation oven at 50 °C and sieved through a 2.0-mm mesh. The following parameters were evaluated for chemical analysis: pH (H₂O) (1:1 ratio), pH SMP (1:1:0.5 ratio), exchangeable aluminum (Al³⁺), calcium (Ca²⁺), and magnesium (Mg²⁺) concentrations (extracted with 1.0 mol L⁻¹ KCl, analyzed through atomic emission spectroscopy), and available P and K (extracted with Mehlich-1, analyzed using the method of Murphy and Riley (1962) for P and atomic emission spectroscopy for K) (Tedesco et al. 1995). Total organic carbon (TOC) was determined using the method of Embrapa (2017) and converted to soil organic matter (SOM) by multiplying by 1.724. Potential acidity (H+Al) was estimated according to Eq. 1. Eqs. 2 and 3 were used to determine the effective cation exchange capacity (CEC_{eff}) and the cation exchange capacity at pH 7.0 (CEC_{pH7.0}). The saturation of CEC_{pH7.0} by Ca-Mg-K was calculated using Eq. 4 (CQFS-RS/SC 2016).

$$H + Al(\text{cmol}_c\text{dm}^{-3}) = e^{(10.665 - (1.1483 \text{ SMP}))} / 10 \quad (1)$$

$$CEC_{eff}(\text{cmol}_c\text{dm}^{-3}) = Ca^{2+} + Mg^{2+} + K^{+} + Al^{3+} \quad (2)$$

$$CEC_{pH7.0}(\text{cmol}_c\text{dm}^{-3}) = Ca^{2+} + Mg^{2+} + K^{+} + (H + Al) \quad (3)$$

$$Ca - Mg - K \text{ saturation}(\%) = ((Ca + Mg + K) / CEC_{pH7.0}) \times 100 \quad (4)$$

2.5 Statistical analysis

Data regarding surface runoff and element loss were analyzed using the Shapiro–Wilk test and O'Neill and Mathews test to verify normality and homogeneity, respectively. When necessary, to meet the assumptions of normality and homoscedasticity, the data were subjected to Box–Cox transformation. The effects of soil management, year, and their interaction were evaluated using two-way analysis of variance (ANOVA). Mean separations were performed using Tukey's test at a 5% significance level ($p \leq 0.05$). These statistical analyses were performed using the R statistical environment (R Core Team 2025).

3 Results

3.1 Soil chemical modification

According to CQFS-RS/SC (2016), the experimental area was characterized as having low-acid soil, with a medium SOM content (2.5%–5.0%) and a high $\text{CEC}_{\text{pH}7.0}$ (15.1–30.0 $\text{cmol}_c \text{ dm}^{-3}$). Following the cultivation and fertilization of common beans, winter cover crops, and garlic in 2023, available P concentrations in the 0.00–0.10 m soil layer were up to 99% higher in soil cultivated with black oat than in soil cultivated with winter pea or forage radish, and that in areas under the CT system (Table 1). In the same layer, the concentration of K was 42% higher under the NT system with cover crops than under the CT system. In the 0.10–0.20-m layer, no differences were observed in soil chemical parameters across treatments in 2023. In 2024, the lowest K concentrations were observed in the winter pea system across both the 0.00–0.10-m and 0.10–0.20-m layers, with an average reduction of 61% relative to black oat. Compared with the other treatments, the cultivation of winter peas also reduced the P concentration in the 0.10–0.20-m soil layer. No effects were observed for other soil chemical parameters evaluated.

3.2 Climate conditions and water, soil, and nutrient losses

The cumulative precipitation was 2,331 mm in 2023 and 1,885 mm in 2024 (Fig. 3). In 2023, 20 rainfall events with surface runoff were recorded; however, in 2024, only 9 events generated runoff because of lower rainfall indices. Between October and November 2023 alone (Fig. 3a), the rainfall volume represented 36% of the total cumulative rainfall for the year. This volume was more than double the average volume expected for these months (Wrege et al. 2012). During these two months (October and November), the most erosive rainfall occurred (Fig. 4). In 2024, the highest accumulated rainfall was recorded in January and May, featuring events of higher rainfall erosivity. From the second half of 2024 onward, fewer intense rainfall events and consequently lower surface runoff were noted.

The soil losses were reduced by approximately 46% when the soil was covered by cover crops (NT) compared with the soil under the CT system (Fig. 5b), regardless of the year evaluated. Water losses by surface runoff were 127% higher in 2023 than in 2024 (Fig. 5a), with no differences noted between cultivation systems. Regardless of the management system, N_T , TOC, and P_{SOL} losses were 84%,

Table 1 Soil chemical parameters in the 0.00–0.10-m and 0.10–0.20-m soil layers after common bean cultivation, followed by winter cover crops (black oat - BO, forage radish - FR, and winter pea - WP) and garlic in the 2023 and 2024 growing seasons

Systems	pH-H ₂ O	SOM	H+Al	Al ³⁺	Ca ²⁺	Mg ²⁺	P	K	V	CEC _{pH7.0}	CEC _{eff}
		%		cmol _c dm ⁻³			mg kg ⁻¹		%	cmol _c dm ⁻³	
2023 crop season											
0.00–0.10-m											
CT	5.1±0.5	4.3±0.5	6.8±3.1	0.5±0.4	3.5±1.9	3.8±1.3	25.3±14.6 b	401.5±143.7 b	45.4±19.0	18.3±3.5	8.8±3.1
BO	5.4±0.4	4.6±0.3	5.6±0.5	0.8±0.3	4.8±2.7	4.0±1.3	56.8±16.8 a	589.5±91.8 a	57.8±12.6	18.2±3.4	11.3±3.2
FR	5.5±0.2	3.8±0.9	5.5±2.6	0.6±0.3	3.0±0.6	3.7±1.4	35.1±5.5 b	590.3±130.9 a	37.8±7.5	15.3±2.2	8.4±1.0
WP	4.9±0.4	4.0±0.2	8.8±0.9	1.3±0.5	2.0±1.3	3.1±1.4	25.1±20.1 b	534.6±88.8 a	29.3±15.9	19.7±4.9	7.1±1.5
0.10–0.20-m											
CT	5.5±0.4	4.5±0.4	5.0±0.0	0.2±0.3	4.7±1.9	4.7±1.0	30.8±10.7	365.5±152.0	60.6±12.0	17.3±2.3	10.7±1.7
BO	5.4±0.1	4.7±0.6	6.6±0.8	0.5±0.4	4.5±1.4	4.7±1.1	38.0±15.5	375.0±169.8	51.9±18.1	20.3±0.5	11.1±0.9
FR	5.8±0.3	4.4±0.8	6.5±1.5	0.0±0.0	3.5±1.0	3.7±0.3	33.7±7.6	379.8±161.7	47.8±9.4	17.7±0.6	8.5±1.5
WP	5.1±0.7	3.8±0.4	7.8±1.8	0.9±0.9	1.9±0.6	3.3±0.8	25.7±14.9	340.0±49.4	32.6±4.3	17.6±2.5	6.6±0.1
2024 crop season											
0.00–0.10-m											
CT	5.5±0.4	3.2±1.1	10.0±5.3	0.6±0.5	3.4±1.0	5.3±1.6	27.7±9.3	365.8±151.7 b	53.4±11.9	19.4±2.4	11.0±1.9
BO	5.7±0.3	3.0±0.6	7.7±1.9	0.2±0.1	4.3±0.2	6.4±0.8	27.2±11.9	659.8±96.0 a	66.1±2.8	19.1±1.4	12.8±0.8
FR	5.8±0.4	2.6±0.4	7.5±0.8	0.3±0.4	4.3±1.3	6.8±1.5	23.1±5.1	432.2±208.2 ab	62.3±21.2	20.2±3.9	12.9±0.3
WP	5.6±0.3	3.2±0.8	13.9±7.2	0.2±0.1	4.0±0.6	6.2±2.0	23.3±15.9	261.3±57.6 c	61.8±10.7	18.7±0.8	11.8±1.9
0.10–0.20-m											
CT	5.2±0.1	2.6±0.3	6.8±3.1	0.5±0.4	3.4±0.9	5.3±1.7	25.2±9.2 a	426.2±178.7 a	54.2±14.6	17.9±3.5	10.2±2.1
BO	5.2±0.3	3.0±1.2	9.8±6.5	1.0±0.7	2.6±0.7	4.7±1.4	21.0±11.7 a	535.2±131.7 a	53.4±10.6	15.5±1.5	9.3±1.2
FR	5.1±0.3	2.3±0.4	9.3±2.3	1.0±1.0	2.6±1.0	5.0±1.9	13.8±8.2 a	434.4±268.8 a	53.3±22.2	16.2±1.0	9.6±2.2
WP	5.0±0.2	2.3±0.5	11.8±4.2	1.6±0.9	2.1±0.7	3.9±1.3	8.7±3.2 b	206.4±54.6 b	39.4±10.1	17.3±2.2	8.5±1.4

CT conventional tillage, SOM soil organic matter, H+Al potential acidity, V Ca-Mg-K saturation, CEC_{eff} effective CEC. Means followed by the same letter in the same column do not differ from each other according to the Tukey's test ($p \leq 0.05$)

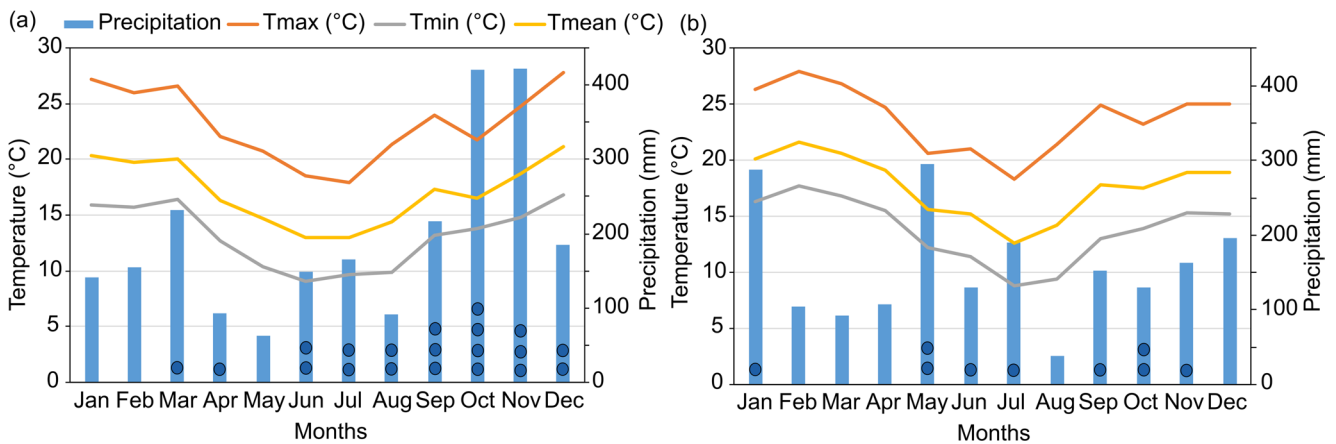
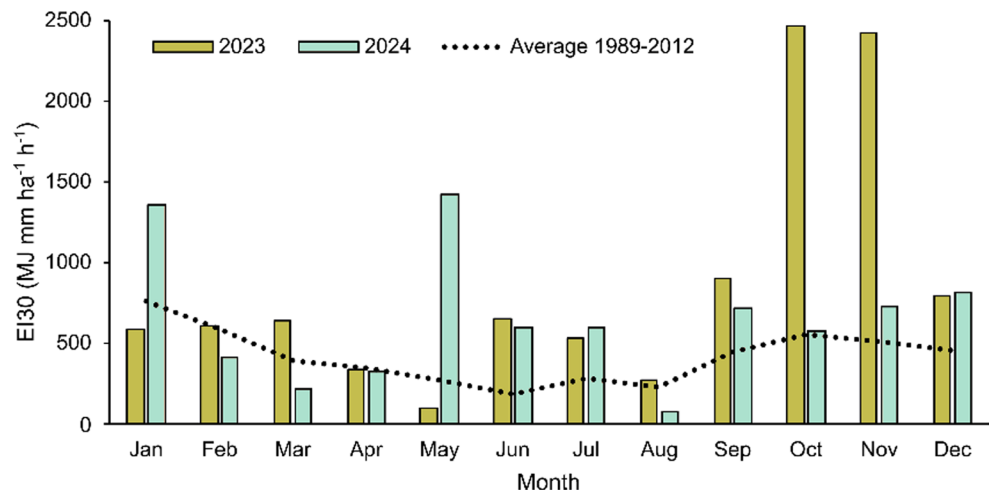


Fig. 3 Monthly precipitation and air temperatures during the 2023 and 2024 agricultural years. The blue dots represent rainfall events with surface runoff

Fig. 4 Monthly values of the EI30 erosivity index in Curitiba-banos, Santa Catarina state. The values were calculated according to Loureiro and Coutinho (2001). Historical averages between 1989 and 2012 were obtained according to the methods of Schick et al. (2014) for Lages, SC



21%, and 155% higher, respectively, in 2023 than in 2024 (Figs. 5c, d, e). No differences in P_T losses (327 g ha^{-1}) were observed for the different cultivation systems or years (Fig. 5f).

3.3 Plant contributions to nutrient loss

The greatest losses of water, soil, and nutrients were observed during garlic cultivation between July and December 2023 (Fig. 6). On average, these values represented 94%, 6%, and 1% of the total losses for garlic, beans, and cover crops, respectively. During the 2024 growing season, the greatest losses of water, soil, and nutrients occurred during the bean cultivation period (January to March), surpassing the values recorded for the same crop in 2023. These values represented an average of 75%, 14%, and 11% of the losses for beans, cover crops, and garlic, respectively.

The N_T concentration in the surface runoff was 15% higher in 2023 (20.2 mg L^{-1}) than in 2024 (Fig. 7a). No differences were observed in the average TOC concentration,

which was 90.7 mg L^{-1} across both years (Fig. 7b). The P_{SOL} concentration was 95% higher in the soil covered with forage radish (1.2 mg L^{-1}) than in the soil under the CT system, although the values did not differ among the soil covered with other cover crops (Fig. 7c). Finally, a 199% increase in the P_T concentration was observed in 2024 (9.9 mg L^{-1}) compared with the 2023 growing season (Fig. 7d).

4 Discussion

4.1 Water and soil losses

The higher losses of water, soil, and nutrients in 2023 were associated with the intensity of rainfall during the growing season, particularly in October and November (Figs. 3 and 4). In the second half of 2023 and the first quarter of 2024, the El Niño phenomenon influenced the region's climate, resulting in intense and concentrated precipitation during spring and summer (Clarke et al. 2026), a typical characteristic of southern

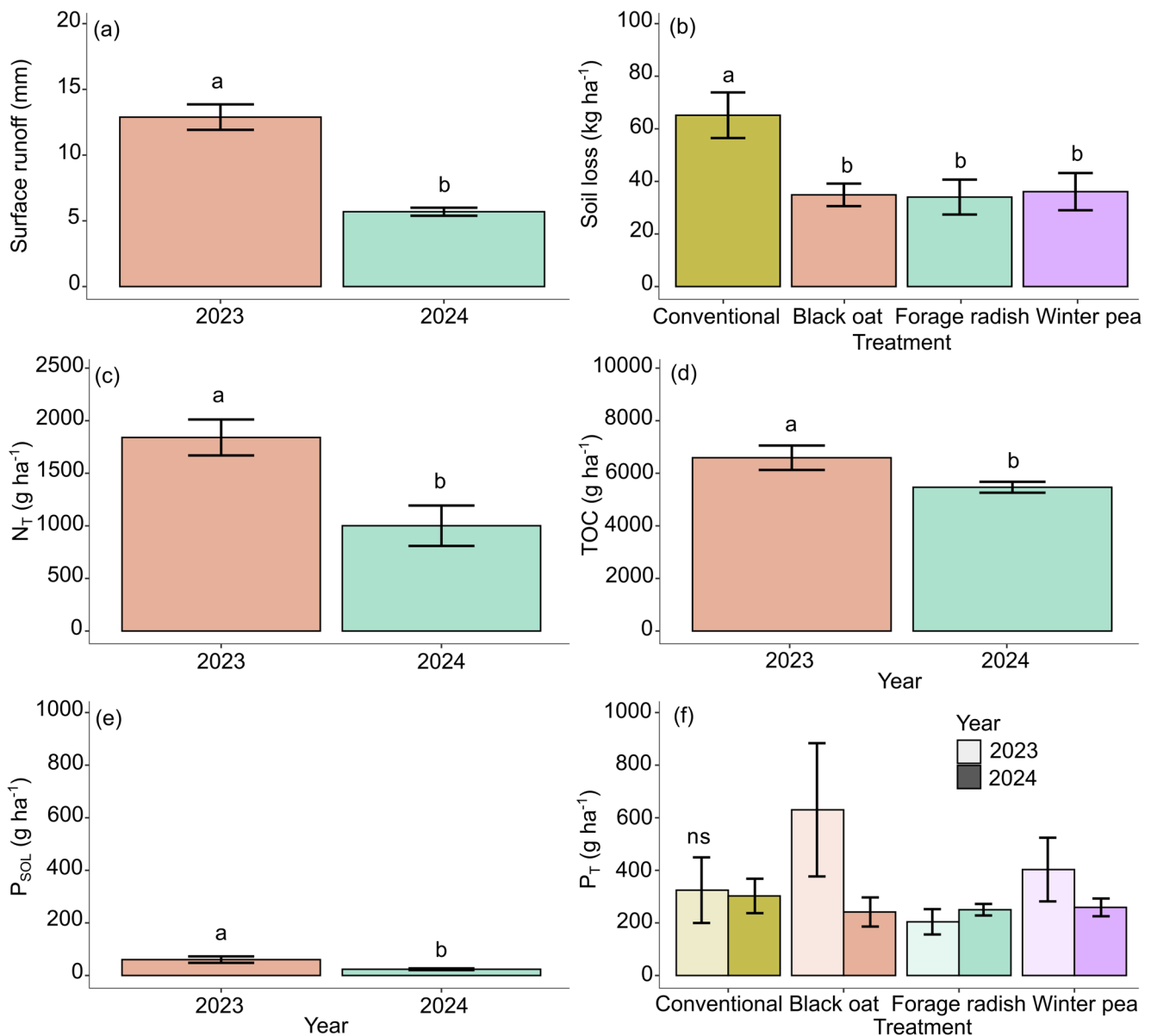


Fig. 5 Water (a), soil (b), total N (N_T) (c), total organic carbon (TOC) (d), soluble P (P_{SOL}) (e), and total P (P_T) (f) losses across different garlic cultivation systems during the 2023 and 2024 growing seasons.

Means followed by the same letter for year or treatment do not differ according to the Tukey's test ($p \leq 0.05$); ns: nonsignificant

Brazil, which is characterized by high rainfall erosivity (Schick et al. 2014). High-intensity rainfall promotes losses, particularly in areas with steeper slopes, and is a determining factor in the magnitude of these losses (Grando et al. 2023).

The sampling units were installed in soil on a 13% slope, which is less steep than the regional average of 22% (Martini 2012). Although the observed water loss values were moderate, they indicate a potential risk of intensified runoff in sloping areas under the CT system, particularly in years with higher rainfall volumes and intensities. Regional literature based on historical series in southern Brazil have reported annual surface runoff losses of 246–313 mm under

the NT system and 403–464 mm under the CT system, with losses reaching 519 mm in bare soil (Dambros et al. 2024). Therefore, complementary soil conservation practices are necessary for controlling concentrated flow and preventing the reduced effectiveness of residue cover over long slope lengths (Barbosa et al. 2021; Du et al. 2022). One practice that requires reconsideration is the construction of beds parallel to the slope to facilitate machinery movement and irrigation system installation. However, this practice does not reduce slope length; instead, it directs runoff water at higher velocities toward lower elevations, intensifying the erosion process (Barbosa et al. 2021).

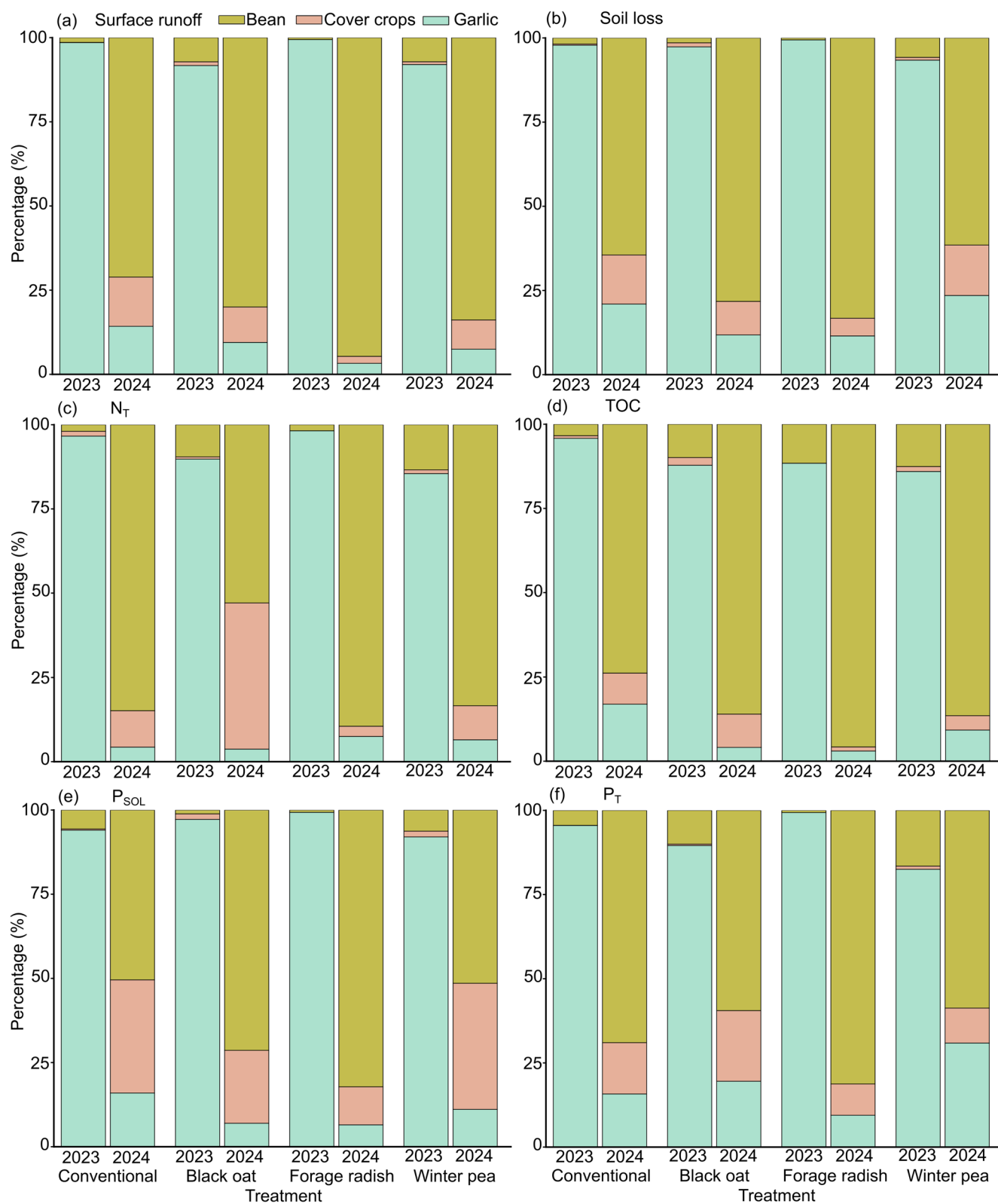


Fig. 6 Percentage losses of water (a), soil (b), total N (N_T) (c), total organic carbon (TOC) (d), soluble P (P_{SOL}) (e), and total P (P_T) (f) across different garlic cultivation systems in 2023 and 2024

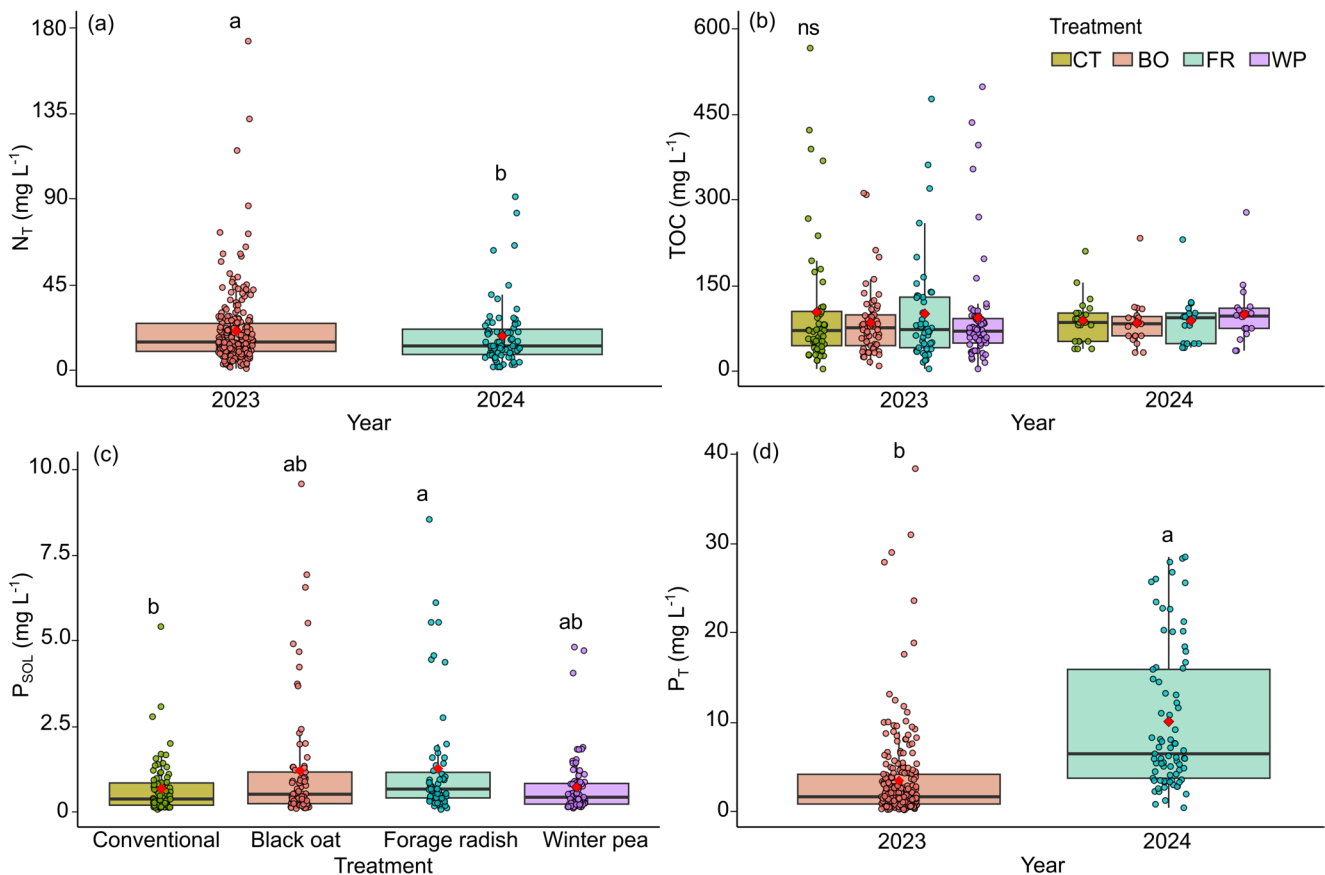


Fig. 7 Concentrations of total N (N_T) (a), total organic carbon (TOC) (b), soluble P (P_{SOL}) (c), and total P (P_T) (d) in the runoff across different garlic cultivation systems in 2023 and 2024. CT: conventional tillage with bed formation and no cover crop prior to garlic cultivation, BO: black oat (*Avena strigosa*) straw, FR: forage radish (*Raphanus sativus* L.) straw, WP: winter pea (*Pisum sativum* L.) straw. The red dots indicate the mean values, whereas the horizontal line inside

the box represents the median. The upper and lower limits of the box represent the 75th and 25th percentiles, respectively. The whiskers (vertical lines) extend to the maximum and minimum values within 1.5 times the interquartile range. The open circles represent individual data points, including potential outliers. Means followed by the same letter for year or treatment do not differ from each other according to the Tukey's test ($p \leq 0.05$); ns: nonsignificant

Although soil loss is often linked to rainfall intensity and volume, losses are exacerbated in areas under the CT system (Carretta et al. 2021; Dambros et al. 2024). Similar data were obtained in our study, where soil losses were higher under the CT system than under the NT system, but no differences were observed between the two evaluated years (Fig. 5b). This is because the CT system disrupts the soil surface, increasing its susceptibility to erosion (Wolschick et al. 2021). Under NT systems, aggregate stability also depends on the rate at which organic matter decomposes, which is associated with the interactions among soil type, moisture, and agricultural management (Blanco-Canqui and Lal 2008). The soil used in the present study had high clay and organic matter contents, which increase soil aggregate stability and improve physical quality and water infiltration (Soinne et al. 2023). The greater protection afforded to organic matter by clay minerals (Xu et al. 2022) results in higher resistance to erosion.

This behavior reflects the effects of the garlic crop and soil characteristics during garlic cultivation, particularly with regard to exposure to surface runoff during periods of high precipitation. Moreover, even if soil preparation is restricted to the planting row, it causes physical disturbance, making the soil more susceptible to losses (Hahn et al. 2024b). Compared with that under a conservation system, the impact of soil loss under a CT system is considerably higher, especially when the frequency of short and intense rainfall events increases erosion (Carretta et al. 2021).

4.2 Nitrogen and phosphorus losses

Given that vegetables such as garlic have high nutritional requirements over a short period, the use of sources with high nutrient availability is recommended. Applying high rates aligns with regional recommendations for this crop, which indicates that 269 to 307 kg ha⁻¹ of N and 397 kg ha⁻¹ of

P_2O_5 are needed to achieve high yields (Hahn et al. 2020, 2024a). In 2023, black oat, forage radish, and winter pea produced 3.0, 3.7, and 2.5 $Mg\ ha^{-1}$ dry matter, respectively (data not shown). In 2024, the biomass of these species was 3.8, 7.1, and 4.0 $Mg\ ha^{-1}$, respectively, indicating an increase in productivity, particularly for forage radish. This increase can be attributed to the higher rainfall volume in May 2024 (Fig. 3b). Notably, P concentrations were above the critical level for garlic and other regional crops (CQFS-RS/SC 2016). Consequently, no changes in plant performance were anticipated, although higher nutritional values in leaf tissue were evident because of luxury consumption (Kang et al. 2014).

Modifications in soil P availability were observed across the different management systems (Table 1). The high P concentrations were related to fertilization, totaling 635 $kg\ ha^{-1}$ of P_2O_5 over the two years. Although a fertilizer was incorporated into the soil under the CT system, no differences in P availability were observed at the 0.10–0.20-m depth were observed compared with cover crop systems, where fertilizer application was restricted to the planting line. The higher P availability observed in the black oat treatments than in the treatments with other cover crops and even in the CT treatment (Table 1) suggests that the lower biomass production requirements and the efficiency of nutrient cycling in this species played a decisive role. Black oat has a robust root system that releases exudates (De Conti et al. 2018), which can solubilize less available forms of P, thus promoting organic matter mineralization and increasing P availability. A reduction in P concentrations in the 0.10–0.20-m layer occurred when the soil was cultivated with winter peas. This can be explained by the higher capacity of winter pea to extract P, which reaches values of 2.0 $kg\ Mg^{-1}$ of P in dry matter compared with 0.9 $kg\ Mg^{-1}$ in black oat (Freund et al. 2021). Increased decomposition rates increase the concentration of organic acids that act in acidification, chelation, and exchange reactions, thereby affecting nutrient dynamics and soil availability (Adeleke et al. 2017).

No differences were observed in the P_{SOL} and P_T losses (Figs. 5e, f) in surface runoff collected from areas under the CT system compared with areas cultivated with cover crops. This occurs because cover crops do not have greater source P export than soil does (Cober et al. 2019), with a direct relationship noted between P losses and soil P concentrations (Grando et al. 2021). Withers et al. (2026) recently revealed that ‘P legacy’ in the soil was the main pollution threat from agriculture, highlighting the significance of reducing soil P reserves to lower waterway P pollution. Legacy P refers to recalcitrant forms associated with clay minerals that can buffer and, at some point, become available, thereby causing environmental impacts (Withers et al. 2018). Therefore, even when cultivated crops are under an NT system, other management practices must be adopted to minimize P losses and conserve water quality (Barbosa et al. 2021; Langhans et al. 2022).

4.3 Impact of P transport on water bodies

The losses of P_{SOL} and P_T did not differ between cultivation years or soil management systems with cover crops (Figs. 5e, f). This occurred because P accumulates in the surface soil layers under the NT system due to the lack of soil disturbance, which increases its availability for surface runoff during intense rain events (Lourenzi et al. 2015). Additionally, P mobility in the soil depends on biomass decomposition dynamics and the diameter of the transported clay fraction (Figueiredo et al. 2025; Sharpley et al. 2015). P transport is intensified by increased water erosion, whereby elevated soil transport leads to greater losses of the P_T fraction (Grando et al. 2021).

In the first crop, P fertilization was performed by applying 435 $kg\ ha^{-1}$ of P_2O_5 to the garlic planting row at a depth of approximately 0.05 m. In contrast, under the CT system, where P was incorporated into the soil, P was diluted within the 0.00–0.20-m layer, which may have contributed to reduced P_{SOL} losses through runoff (Fig. 7c). Although the observed nutrient losses may seem small, they can result in concentrations above the limits established by Brazilian legislation when transported to water bodies. This legislation establishes a P_T limit of 0.15 $mg\ L^{-1}$ for the most lenient water quality class (BRAZIL 2005). These limits are also higher than the critical levels for eutrophication, which are 0.075 $mg\ L^{-1}$ in streams (Dodds et al. 1998) and 0.064 $mg\ L^{-1}$ in reservoirs (Cunha et al. 2013). Therefore, P_T losses such as those recorded in this study—22 and 66 times higher than the limits established by Brazilian legislation in the 2023 and 2024 evaluations—could represent a potential risk of eutrophication if mitigating practices are not adopted. The higher P_T concentration in 2024 than in 2023 was associated with lower water losses through surface runoff.

The effectiveness of cover crops in terms of P retention is directly related to their ability to form a dense and persistent cover throughout the agricultural cycle. Our results emphasize the importance of considering not only the management system but also the characteristics of the cover crop species in P management. Furthermore, cover crops can reduce the need for fertilization (Hahn et al. 2024b). However, additional studies are warranted to evaluate the response of these crops to reduced fertilizer rates under increasing soil cover rates.

4.4 Management indications

This study revealed that, from the standpoint of soil and water conservation, CT is not a sustainable practice in horticultural systems. Therefore, the use of cover crops prior to garlic cultivation is recommended to increase soil protection and minimize erosion. However, cover crops appeared to

be more effective in years with lower rainfall. Therefore, to reduce slope length and the effects of the erosive processes, it is advisable to avoid steep slopes and/or soil disturbance for bed formation along the slope direction.

5 Conclusion

Water, soil, and nutrient losses are dependent on precipitation and the erosivity index. Compared with the CT system, the NT system reduced soil loss in the horticultural system, which demonstrates the importance of soil protection in preventing erosive processes. The losses of N_T , TOC, and P_{SOL} were more significant in the year with higher rainfall, with no effects of the cultivation system, indicating that their dynamics may vary according to the climatic conditions of each growing season. The lowest percentages of water, soil, and nutrient losses occurred when the soil was cultivated with cover crops rather than with garlic or bean crops. The P_T concentrations in the surface runoff exceeded the limits established by environmental legislation, indicating the importance of controlling surface runoff in horticultural systems.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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